

Problem Set 2

Metrics spaces and bases

Solve exercise 6 and **four** of the other five problems.

Problem 1

Let (X, d) be a metric space, and define a new metric on X by

$$\bar{d}(x, y) := \frac{d(x, y)}{1 + d(x, y)}.$$

1. Verify that this is a metric.
2. Verify that d and $\bar{d}$ induce the same topology on X .
3. For $X = \mathbb{R}$, prove that d and $\bar{d}$ are not equivalent in general.

Problem 2

Let (X, d) be a metric space and let $Y \subset X$ be a subset. The restriction of the distance d to Y defines a metric d_Y and thus a topology $\mathcal{T}_{d_Y}$ on Y . Verify that this topology coincides with the topology on $Y \subset X$ induced by the topology $\mathcal{T}_d$ on X .

Problem 3

1. Prove that the set of open intervals $\mathcal{B} = \{(a, b) \mid a < b\}$ is a basis that generates the usual topology on $\mathbb{R}$.
2. Prove that the set of half-open intervals $\mathcal{B}_\ell = \{[a, b) \mid a < b\}$ forms a basis for a topology on the set $X = \mathbb{R}$. Compare the resulting topological space $\mathbb{R}_\ell$ with the standard topology on $\mathbb{R}$, i.e. which topology is finer/coarser? ¹
3. For a set X with at least two elements, prove that $S = \{X \setminus \{x\} \mid x \in X\}$ is a subbasis for the cofinite topology on X (see the third exercise of p-set 1 for the definition of the cofinite topology).

Problem 4

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1. To compare these two topologies, you may use the first item of Problem 5.

1. Let X, Y be spaces, and let $\mathcal{B}$ be a basis for the topology on X . Prove that a map $f: X \rightarrow Y$ is open if and only if $f(B) \subset Y$ is open for every $B \in \mathcal{B}$.
2. Prove that a map $f: X \rightarrow Y$ is continuous for a topology $\mathcal{T}_S$ on Y if and only if $f^{-1}(U) \subset X$ is open for every $U \in S$.²
3. Let $\mathcal{B} = \{B_i\}_i$ be a basis for a topology on X , and let $f: X \rightarrow Y$ be a continuous open surjective function. Prove that $f(\mathcal{B}) = \{f(B_i)\}_{i \in I}$ is a basis for the topology on Y .

Problem 5

The goal of this exercise is to show that if $\mathbb{R}$ is endowed with the standard topology, then the box topology on $\mathbb{R}^n$ coincides with the standard topology.

1. If $\mathcal{B}$ and $\mathcal{B}'$ are two bases on a set X , then $\mathcal{T}_{\mathcal{B}'}$ is finer than $\mathcal{T}_{\mathcal{B}}$ if and only if for every $x \in X$ and every $B \in \mathcal{B}$ that contains x , there exists a $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.
2. Prove that the open balls for d_∞ on $\mathbb{R}^n$ are

$$B_{d_\infty}(x, r) = (x_1 - r, x_1 + r) \times (x_2 - r, x_2 + r) \times \dots \times (x_n - r, x_n + r).$$

3. Prove that the box topology on $\mathbb{R}^n$ coincides with the standard topology.

Problem 6

Given two groups G, H , a map $f: G \rightarrow H$ is called a *group homomorphism* if $f(gg') = f(g)f(g')$ for every $g, g' \in G$.

1. Prove that $f(e_G) = e_H$ and $f(g^{-1}) = f(g)^{-1}$ for every $g \in G$.
2. Prove that $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ is a group under complex multiplication $\cdot$, then prove that the exponential map defines a group homomorphism $(\mathbb{R}, +) \rightarrow (S^1, \cdot)$.
3. Prove that $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ is a group under addition modulo n . How many group homomorphisms $f: \mathbb{Z}_n \rightarrow \mathbb{Z}_m$ are there?

2. Here, it is understood that $S \subset \mathcal{P}(Y)$ is a subbasis.